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Accurate surface and adsorption energies from many-body perturbation theory

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Kohn–Sham density functional theory is the workhorse computational method in materials and surface science[1]. Unfortunately, most semilocal density functionals predict surfaces to be more stable than they are experimentally. Naively, we would expect that consequently adsorption energies on surfaces are too small as well, but the contrary is often found: chemisorption energies are usually overestimated[2]. Modifying the functional improves either the adsorption energy or the surface energy but always worsens the other aspect. This suggests that semilocal density functionals possess a fundamental flaw that is difficult to cure, and alternative methods are urgently needed. Here we show that a computationally fairly efficient many-electron approach, the random phase approximation[3] to the correlation energy, resolves this dilemma and yields at the same time excellent lattice constants, surface energies and adsorption energies for carbon monoxide and benzene on transition-metal surfaces.

The random phase approximation (RPA) predicts not only very good adsorption energies but also the correct carbon monoxide (CO) adsorption site on all considered substrates, resolving the second puzzle that was first noticed in 2001 by Feibelman and co-workers in their now classic paper[4]. They observed that no density functional was able to predict the correct adsorption site for CO on Pt(111). Since then many different variants of Kohn–Sham density functional theory (DFT; ref. 1) have been applied to this and similar prototypical problems, and, although there is now evidence that the origin of the problem is an overestimation of the bond strength of the empty molecular frontier orbital with the substrate[5–8], parameter-free calculations have been unable to yield predictions in agreement with experiment[8].

Although the site-preference problem is considered to be a significant issue, the general tendency of semilocal density functionals to underestimate surface energies and to overestimate chemisorption energies is even more troublesome, because it makes reliable predictions of surface reconstructions challenging. Too small surface energies and too large adsorption energies are also counterintuitive, because a small surface energy implies too large stability, which should result in too small adsorption energies. However, the contrary is found. To exemplify this problem, Fig. 1 shows the calculated surface energies and the calculated adsorption energies for CO on Pt(111) and Rh(111). The most common density functional in materials and surface science is the Perdew–Burke–Ernzerhof (PBE) functional[9], which behaves along the lines discussed above. Semilocal functionals can be tuned by increasing or decreasing the strength of the gradient corrections in the functional[10]. At one extreme, the local density approximation is recovered, predicting reasonable surface energies but vastly overestimating adsorption energies. Close to this extreme is the semilocal PBEsol (ref. 12) functional, whereas a functional representative of very strong gradient corrections is the Becke–Lee–Yang–Parr (BLYP) functional[13]. The important, but not unexpected, observation is that all functionals lie roughly on a straight line: with increasing surface energy the absolute adsorption energy increases as well, and vice versa. However, the theoretical line is clearly offset from the experimental values, making quantitative predictions for surface-science problems difficult. As a possible solution to the problem, hybrid functionals that admix a fixed fraction of the non-local Hartree–Fock exchange were considered, but obviously they do not solve the dilemma either (see Fig. 1b, points HSE; ref. 8).

When exact Hartree–Fock exchange is combined with the RPA to the correlation energy[3], immediate improvements compared with semilocal DFT calculations are observed (blue points in Fig. 1b). Starting from PBE orbitals, RPA increases the surface energy, but decreases the adsorption energy, yielding very good agreement with experiment. To gain insight into this behaviour we have evaluated the quasiparticle energies using a method that is compatible with RPA total-energy calculations, the so-called GW method in the RPA approximation (GW–RPA; ref. 14). This method enables direct comparison with experimental photoemission or inverse-photoemission measurements. The electronic densities of states (DOSs) for a semilocal DFT functional (PBE), for GW–RPA and for a hybrid functional (HSE) are shown in Fig. 2. The common model to describe CO adsorption is the Blyholder model[15,16], which invokes interactions of the two CO frontier orbitals, the 5σ highest occupied molecular orbital and the $2\pi^*$ lowest unoccupied molecular orbital, with the metal states. Because of the interaction with the metal states, bonding and antibonding 5σ–metal hybrid orbitals develop, and the latter are partly shifted above the Fermi level of the metal, causing a net bonding interaction (donation to the surface). Likewise, bonding $2\pi^*$–metal hybrid states become populated (back-donation). From symmetry arguments it follows that the highly directional 5σ–metal interaction is particularly strong for atop adsorption, whereas the $2\pi^*$ interaction is more important for hollow-site adsorption[5,6,17].

It is well established that semilocal functionals predict a very reasonable d-band width for Pt. Indeed, comparison of the DFT results with GW–RPA, and comparison of the bulk DOS with experimental spectra, confirms this. However, PBE fails to predict the position of the CO frontier orbitals. Both the 5σ and, in particular, the $2\pi^*$ orbitals are located too close to the Fermi level, which we believe to cause the observed overestimation of the adsorption energies[5]. On modification of the semilocal density functional, the DOS remains essentially unchanged, but the surface energies and adsorption energies increase or decrease, observing a linear relationship. As the molecular frontier orbitals are too

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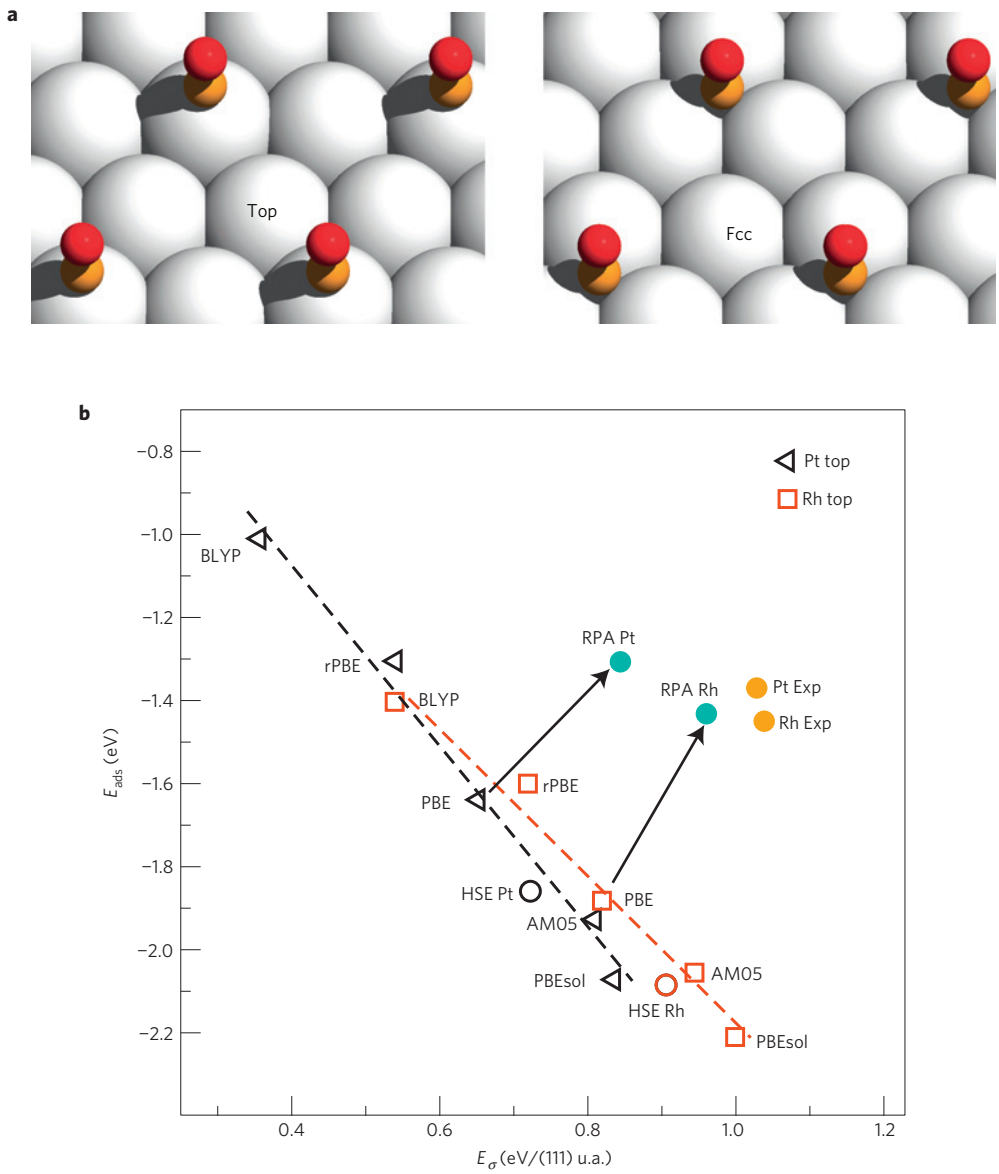

Figure 1 | Atop CO adsorption and surface energies for Pt(111) and Rh(111). a, Considered CO adsorption geometries for a (2 × 2) surface cell. Semilocal functionals predict CO to adsorb in the fcc hollow site coordinated to three metal atoms on Pt and Rh, whereas experiments unequivocally show adsorption atop a metal atom. **b**, Atop adsorption energies versus surface energies for Pt(111) and Rh(111); surface energies are specified per unit area (u.a.). Various semilocal functionals were used: AM05 (ref. 11), PBEsol (ref. 12), PBE (ref. 9), rPBE (ref. 2) and BLYP (ref. 13), in order of increasing gradient corrections. Furthermore, the hybrid functional HSE (ref. 18) based on the PBE functional was used.

close to the Fermi level, the adsorption energy is too large if the surface energy is correct. Good agreement with the experimental adsorption energies is only obtained by making surfaces artificially stable using, for instance, the BLYP functional. In this case, however, a better overall description is not obtained either, because BLYP underbinds the solids dramatically and yields for example much overestimated lattice constants (see Fig. 3b). The hybrid functional describes the position of the $2\pi^*$ and 5σ frontier orbitals best, but the bandwidth of the transition metal is overestimated compared with experiment. As this increases the surface energy and concomitantly the capability of the surface to form bonds with adsorbates, the beneficial effects of the hybrid functional on the description of the CO molecule are counteracted by the adverse effects on the description of the substrate. The GW–RPA method yields excellent results for the metal bandwidth and the $2\pi^*$ frontier orbital, but somewhat underbinds the 5σ orbital, which is a known problem of the GW–RPA approximation and commonly observed for localized states (5σ and 1π in our case). Overall, the RPA seems to offer a very reasonable description of the electronic properties (DOS) of CO and the metal, which is also reflected in the improved energetics.

In passing we note that hybrid functionals work well for a closed *d* shell, for example Cu (refs 8,21). In this case, even low-order many-body perturbation theory, such as second-order Møller–Plesset perturbation theory, applied to small clusters[21] yields results in good agreement with cluster-based RPA calculations[22] or periodic RPA calculations[23]. However, open-shell transition metals are more difficult and pose a more stringent test for the accuracy of the RPA.

The adsorption energies for CO at the metal top and hollow sites are summarized in Fig. 3c for Cu, the late 4*d* metals and Pt. Structural optimization was only considered for CO on Rh and Pt. For the top and hexagonal close-packed (hcp) sites, the distance from the surface increases by about 0.03 Å from the PBE

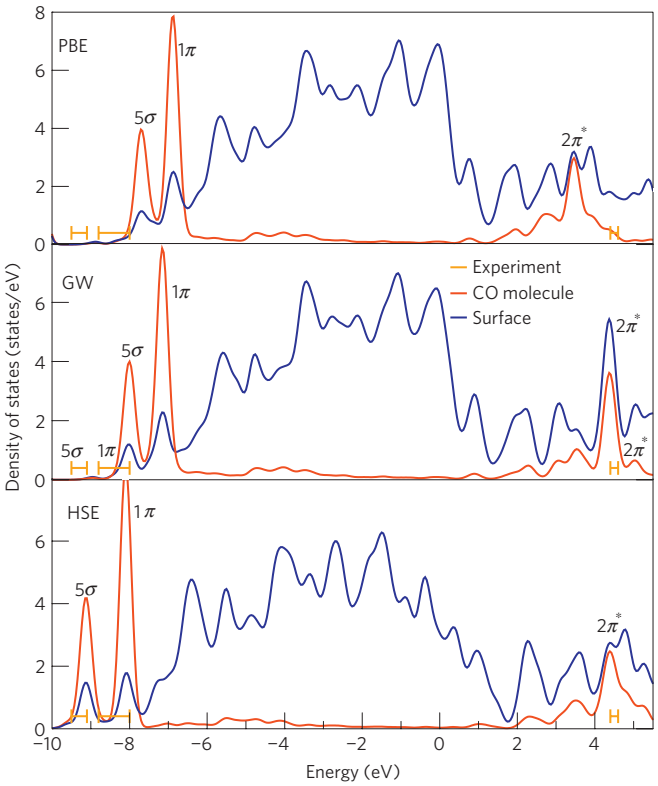

Figure 2 | Electronic DOS for CO adsorbed atop a Pt atom on Pt(111). The DOS is evaluated using DFT (PBE), the RPA (GW) and a hybrid functional (HSE). Experimental photoemission data for the $2\pi^*$ state are from ref. 19, for the 5σ and 1π state from ref. 20.

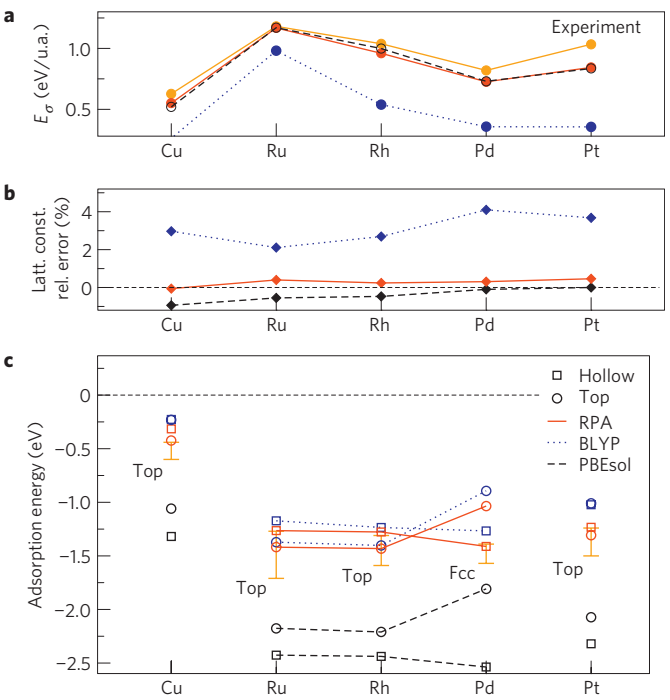

Figure 3 | Surface energies, lattice constants and adsorption energies. **a**, Fcc(111) surface energies (E_σ) for PBEsol, BLYP and RPA. Experimental surface energies are deduced from liquid-metal data[24,25]. **b**, Lattice constants for PBEsol, RPA and BLYP. **c**, Adsorption energies for the atop and hollow sites of CO on Cu, late 4*d* metals and Pt for PBEsol, RPA and BLYP. Experimental data with error bars are from ref. 26. The error bars correspond to the spread of the experimental results.

values, whereas the bond length remains almost identical to the PBE values. The adsorption energies changed by at most 10 meV when the geometries were relaxed using the RPA. Hence, for the other metals, the PBE geometry was used. Excellent agreement with experiment is found for the RPA surface energies, lattice constants and adsorption energies. Furthermore, RPA predicts the correct adsorption site in every single case, but we feel that the correct description of the adsorption energies is even more relevant. The difference between RPA and PBE is a roughly constant shift of the adsorption energies for atop and hollow site adsorption by 350 and 550 meV, respectively. The three most critical cases are Cu, Pt and Rh, where most DFT functionals predict the wrong site order. The RPA restores the correct site order in all cases: -0.42 eV (Cu top) < -0.32 (Cu face-centred cubic (fcc)), -1.31 eV (Pt top) < -1.23 (Pt fcc) and -1.43 eV (Rh top) < -1.28 (Rh hcp).

To reinforce our previous arguments we also show results using the functionals that predict the largest (PBEsol) and smallest (BLYP) surface energy. The functional with weak gradient corrections, PBEsol, yields good lattice constants, and the functional with the largest gradient corrections, BLYP, predicts good adsorption energies. Clearly, however, the BLYP functional (good adsorption energies) predicts much too small surface energies and too large lattice constants, whereas the PBEsol functional (good lattice constants) fails dramatically for the adsorption energies, and only RPA reproduces both.

The improved description within the RPA is not restricted to the adsorption of CO, but holds more generally. An industrially relevant example is the adsorption of aromatic molecules on metal substrates. For close-packed noble-metal surfaces it is widely accepted that the interaction is dominated by van der Waals interactions[27], but on more reactive substrates a mixture between chemisorption and van der Waals bonding is also expected, where the latter is entirely ignored using semilocal functionals[28]. As an example, we have considered the adsorption of a $(\sqrt{7} \times \sqrt{7})$ overlayer of benzene on the ferromagnetic Ni(111) substrate. PBE calculations favour an adsorption in the bridge site by 30 meV, whereas the site order is changed in the RPA calculations, favouring the experimentally observed hollow site by 85 meV. In contrast to the adsorption of CO, the total adsorption energy within the RPA (-0.95 eV) is increased with respect to the PBE calculations (-0.65 eV). This is most likely due to van der Waals contributions, which are entirely ignored for semilocal functionals (PBE), but included in the RPA, as shown before for benzene adsorption on Ag (ref. 27).

In summary, we suggest that the RPA is an excellent choice for predicting surface and adsorption energies. If surfaces reconstruct on adsorption of molecules, a reliable prediction of both properties is absolutely required. At present, this can be only achieved using the methods proposed here. Combined with the improved description of bulk materials[23], the method seems to be an ideal supplement to DFT calculations and an important step towards the accurate and quantitative description of materials properties. It is readily applicable to more applied problems in heterogeneous catalysis or energy storage materials. In contrast to other methods (see for example ref. 10), the RPA does not require fitting to experimental data and is truly parameter free.

Method

The *ab initio* calculations presented here were carried out using the VASP code, PAW (ref. 29) potentials in the implementation of Kresse and Joubert[30] and a cutoff energy of 400 eV. As trends are the main focus of this study, all metals were considered in the fcc structure, although Ru crystallizes in the hcp structure. Adsorption at the atop and fcc hollow sites was considered for Cu, Pd and Pt, whereas for Ru and Rh the top and hcp hollow sites were considered.

Methodological details concerning the RPA calculations, as well as tests for simple prototypical solids, can be found in ref. 23 and references therein. The GW–RPA implementation is discussed in ref. 31. GW–RPA is compatible with RPA total energies insofar as both predict identical electron addition and removal energies.

For the RPA calculations, the surfaces were modelled by a laterally periodic four-layer metal slab with a $(\sqrt{3}\times\sqrt{3})$ periodicity. The DFT and hybrid-functional calculations were carried out using (2×2) supercells and six-layer slabs. For the six-layer DFT calculations, the k-point grid was increased until millielectronvolt convergence was obtained (up to $20\times 20\times 1$ k points). The reported RPA energies were calculated as $E = E_{\mathrm{RPA}}((\sqrt{3}\times\sqrt{3}), 4\,\mathrm{layers}) - E_{\mathrm{DFT}}((\sqrt{3}\times\sqrt{3}), 4\,\mathrm{layers}) + E_{\mathrm{DFT}}((2\times 2), 6\,\mathrm{layers})$. For the four-layer calculations, the same k-point set was used in DFT and RPA, applying at least $6\times 6\times 1$ k points. In selected cases, such as Cu, $10\times 10\times 1$ k points were used. For Cu, Pt and Rh the RPA calculations were also carried out for a (2×2) supercell, confirming that the applied procedure yields adsorption energies accurate to within 30 meV. The adsorption of benzene was calculated in a $(\sqrt{7}\times\sqrt{7})$ supercell using an $8\times 8\times 1$ k-point set.

The RPA calculations were carried out at the PBE ground-state geometries. To evaluate this procedure, the CO bond length and distance from the surface were also optimized using RPA for Pt and Rh as mentioned in the main text. For reasons of consistency, all DFT calculations were carried out at the PBE ground-state geometries, as well. In extreme cases (for example BLYP), the use of the respective relaxed structures would change the energy visibly on the scale of the plots. Nevertheless, this in no way changes the presented trends in adsorption and surface energies.

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Author contributions

L.S., J.H., A.S., F.M. and G.K. carried out the calculations. J.H., A.G., M.M. and G.K. contributed to the implementation of hybrid functionals and RPA. G.K. prepared the manuscript initially. All authors contributed to the discussions and revisions of the manuscript.

Additional information

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